PROJECT COLLECTION

REZA BABAPIRI

Mechanical Design & Manufacturing

CAD, CAM, manufacturing and industrial design practice

2 selected documented projects

Project context / Methods / Evidence / Limitations

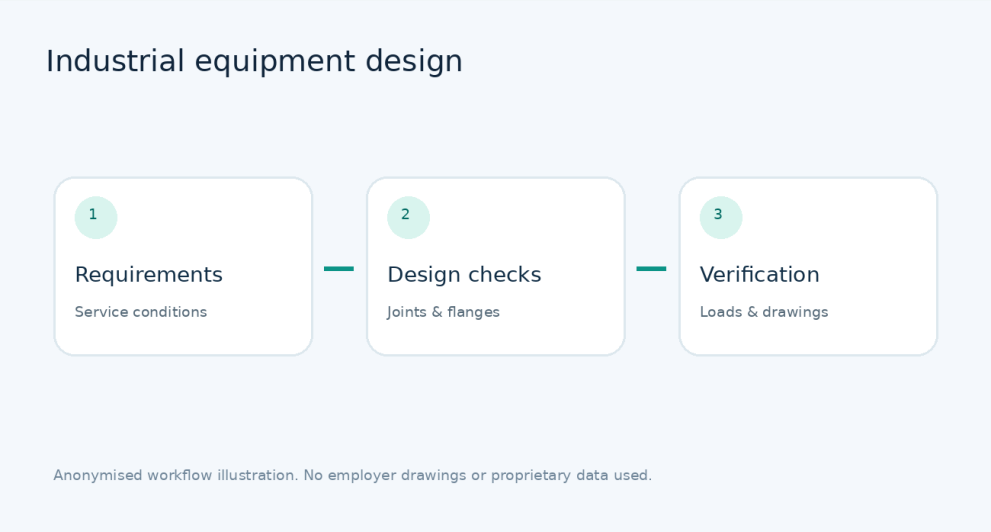
reza-babapiri.github.io

Academic work and anonymised professional experience | Source-derived summaries

PROFESSIONAL EXPERIENCE · INDUSTRIAL EQUIPMENT

Industrial Expansion Joints | Design Practice

Bellows, flanges, piping components and mechanical design verification



ENGINEERING SUMMARY

During industrial mechanical design work at Enetaf Sane Co. (Jan 2022–Jul 2023), worked on expansion joints, bellows, flanges and piping-related equipment. The focus was translating service requirements into manufacturable CAD designs, engineering calculations, design checks and drawing revisions. This is an anonymised professional experience case study, not a reconstruction of a client project.

METHOD / CONTRIBUTIONS

- Produced preliminary AutoCAD layouts and SolidWorks part, assembly and manufacturing drawings.
- Used EJMA-based engineering calculations and applicable ASME and project standards in bellows, material and piping-component design checks.
- Reviewed guide-rod arrangements, bolted connections and flange thickness under operating-pressure and load considerations; used SolidWorks Simulation and ABAQUS for structural assessment.
- Proposed material and thickness revisions to address corrosion concerns and supported selection of pipe and valve specifications for service conditions.

RESULTS & EVIDENCE

- Examples of engineering decisions include material/thickness changes for corrosion mitigation and guide-rod quantity or flange thickness adjustments based on loads.
- Work involved technical document reviews and coordination with fabrication teams.

SCOPE / LIMITATION

Specific client drawings, confidential calculations, certified designs, numerical performance results and test data are not available for public release. Details are based on the engineer’s description of professional experience.

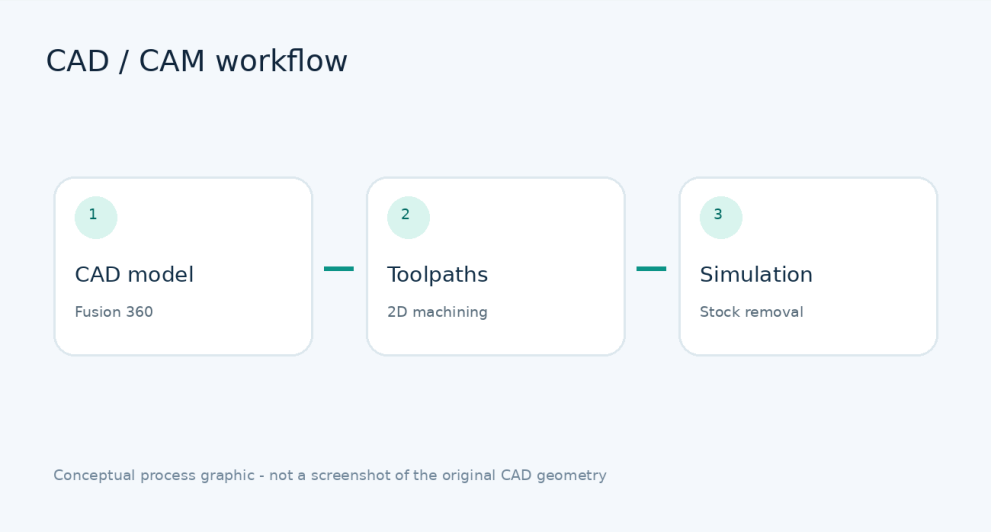
SUPPORTING RECORD

Employment experience described by the author; no confidential employer CAD assets reproduced.

DIGITAL MANUFACTURING · CAD/CAM

Fusion 360 | CAD/CAM CNC Workflow

Aluminium 6061 milling plan with machining simulation and NC output



ENGINEERING SUMMARY

Prepared a Fusion 360 CAD/CAM course challenge for an aluminium 6061 workpiece, setting up machining coordinates and tooling, then creating and validating a multi-operation 2-D milling workflow.

METHOD / CONTRIBUTIONS

- Defined stock, tool library, work coordinate origin and milling setup.
- Created Face, Adaptive Clearing, Pocket, Contour and drilling toolpaths.
- Simulated material removal and exported an NC program using a generic post.

RESULTS & EVIDENCE

- The course summary confirms simulated stock removal and NC export; the Fusion design archive (.f3d) is available.

SCOPE / LIMITATION

A CAM simulation and generic NC post do not establish physical machining quality or actual machine validation.

SUPPORTING RECORD

Fusion .f3d file plus one-page process report.